



Vladislav Nazarenko

Systems Engineer

<https://socket.de/vna>



SOCKET

Key Skills

OpenStack

Ceph

Kubernetes

Qualifications

Experienced Systems Engineer for modern infrastructure with a focus on OpenStack, Ceph, Kubernetes, and Linux. Specialized in the architecture of highly available and scalable cloud platforms, as well as custom solutions for containers and monitoring. Certified in Cloud, Storage, Networking, Monitoring, and Automation.

Fluent in German & English.

Key Information

Focus	Highly available and scalable private cloud platforms
Location	Rhein-Main Region / Remote
Years of Experience	19
Hourly rate	On request

Personal Information	
Name	Vladislav Nazarenko
Education	Taganrog State University of Radio Engineering, Russia
Degree	Engineer in the field of Computer-Aided Design (CAD)
Age	43
Languages	English, German, Russian
Email	vnazarenko@socket.de
Technology Stack	
	OpenStack Ceph S3 MinIO GlusterFS Kubernetes OpenShift Cluster API Gardener Linux RedHat Debian Ubuntu SuSE Networks SONiC Cisco Arista Ansible Puppet Python Ruby C/C++ Shell/Bash PHP PowerShell Prometheus, Grafana, ElasticSearch, OpenSearch, Kibana Artifactory Nexus Quay Harbor MySQL PostgreSQL MongoDB
Certifications	
2024	Prometheus Certified Associate
2023	Red Hat Certified Specialist in Ceph Cloud Storage
2021	Certified OpenStack Administrator
2019	Certified Kubernetes Administrator
2018	Red Hat Certified Specialist in Ansible Automation
2017	Cisco Certified Entry Networking Technician
2014	Puppet Certified Professional
2013	SuSE Certified Linux Administrator
2012	ITIL 2011 Foundation
2011	Red Hat Certified System Administrator
2011	MySQL Professional Database Administrator
Experience	
	Systems Engineer DevOps Engineer Software Engineer Database Administrator
Personal Attributes	
	Hands-on Solution-oriented Reliable

Timeline	Project
10/2022 - PFALZKOM Systems Engineer OpenStack, Docker, Kubernetes, Networks, Ceph, Kolla-Ansible, Baremetal, Artifactory, ElasticSearch, Prometheus	Operational Support and Continuous Development The client has been successfully running a private cloud based on OpenStack for several years. Numerous projects, both large and small, have been collaboratively executed. This ongoing engagement focuses on maintaining the existing OpenStack infrastructure, implementing new feature requirements, and optimizing operational processes. For the deployed OpenStack environments, monitoring solutions leveraging Prometheus and Grafana have been implemented, alongside centralized logging using OpenSearch. Given the continuous evolution of OpenStack and its component services - as well as rising user demands for the private cloud - there is a frequent need to extend OpenStack capabilities and integrate additional services. Recent enhancements include the deployment of Load Balancing (Octavia), DNS (Designate), and Kubernetes (Magnum) services within the OpenStack ecosystem. A notable feature is the Bare Metal provisioning service (Ironic), which, in combination with the Kubernetes service, enables users to provision Kubernetes clusters on bare metal nodes efficiently via the OpenStack dashboard or GUI. Responsibilities: • Scaling and expanding existing OpenStack clusters • Enhancing operational stability and managing release upgrades • Automating workflows and deployment pipelines Tasks: • Integration and deployment of new OpenStack services • Performing regular updates and patch management on clusters • Integrating and managing storage backend systems • Maintaining and updating technical documentation • Diagnosing and resolving bugs • Providing user support and troubleshooting
11/2022 - 08/2023 SINC	Design and Deployment of a Modern Container Platform The objective of the project was to build an open-source CaaS (Container as a Service) platform. The end users should be empowered to provision the necessary infrastructure independently to operate their applications. During the project, numerous components and frameworks were evaluated. The team selected the OSISM distribution (based on

Systems Engineer	OpenStack and Ceph) as the platform foundation. Gardener was used for provisioning and managing Kubernetes clusters.
OpenStack, Ceph, Kubernetes, Gardener, Cluster API, Networks, S3, Kolla-Ansible, Monitoring, Quay, Velero, Helm	Additionally, to ensure a seamless user experience, various tools and technologies were integrated, along with connections to external systems such as DNS, load balancing, observability, logging, and backup & recovery. Throughout the project, involved personnel were trained and prepared for platform operation. Finally, comprehensive documentation was created and handed over to support ongoing maintenance. Requirements • Implementation of a Kubernetes-as-a-Service platform • Open-source based • Support in knowledge transfer and capability building Responsibilities • Platform design and deployment • Configuration management • Integration with external systems • Training and knowledge transfer • Documentation creation
10/2021 - 09/2022	Migration of OpenStack Clusters
PFALZKOM	The client has successfully operated a private cloud based on OpenStack for several years. The objective of this project was to improve the operational efficiency of the OpenStack clusters. Management aimed for a more flexible operation, faster release upgrades, and enhanced scalability.
Systems Engineer	To achieve this, the OpenStack environments were migrated from the traditional setup to containerized services using Kolla-Ansible. At the client's request, the migration was performed without any service interruption, ensuring that end users were unaware of the transition.
OpenStack, Docker, Networks, Ceph, Kolla-Ansible, Artifactory	To improve testability during release upgrades, an automated verification process was implemented to continuously validate cluster functionality and proactively detect regressions or issues following release upgrades or other changes. Requirements • Improved operational efficiency • Accelerated release upgrades • Enhanced scalability

- Zero service downtime
- Improved testability

Responsibilities

- Documentation development
- Cluster rebuild and migration
- Regular updates and maintenance of existing clusters
- Integration with storage backend systems
- User support and bug fixing

12/2020 – 12/2021

OpenShift Administration

Government Authority

The project aimed to establish a modern infrastructure that supports agile methodologies while remaining compatible with traditional approaches, allowing for the gradual migration of existing applications after the platform setup.

DevOps Engineer

**OpenShift, S3 Storage,
ElasticSearch, Ansible,
Nvidia GPU,
Artifactory**

Multiple OpenShift 4.x clusters were deployed, with the primary focus on integrating them into the existing infrastructure components such as load balancers, storage systems, LDAP, image registries, and Ansible Tower.

Concurrently, development was underway on an AI application, which was planned to be operated on the OpenShift platform. To facilitate this, a deployment process was developed, and an Elasticsearch cluster was provisioned to enable a rapid and low-effort migration to OpenShift.

Requirements

- Operational stability
- Support for autonomous team workflows
- Operation of an AI application (both on bare metal and within OpenShift)

Responsibilities

- Administration and maintenance of OpenShift clusters
- Deployment and management of Elasticsearch clusters
- Development and management of deployment pipelines
- Integration of S3-compatible storage systems
- Developer support and troubleshooting

06/2020 – 10/2020

POC: Introduction of OpenShift

Hornbach Baumarkt

In recent years, the customer has deployed several large Kubernetes clusters within their own data centers to enable the operation of containerized applications.

DevOps Engineer

However, new operational challenges have emerged in managing these

**OpenShift, OKD,
OpenStack,
Kubernetes, CoreOS**

Kubernetes clusters, increasing the workload on the operations team. Additionally, the application development teams expressed a desire for more agility in scaling Kubernetes clusters up and down.

The project aimed to assess the feasibility of replacing the existing Kubernetes clusters with OpenShift clusters, thereby improving operational support and reducing the burden on the operations team.

Requirements

- Automated installation and updates of clusters
- Support for scale and auto-scale capabilities
- Cluster homogeneity
- Extensibility through integration with operatorhub.io
- Support for operational workflows

Responsibilities

- Evaluation of OpenShift 4.x and OKD 4.5 versions
- Preparation for the replacement of existing Kubernetes clusters
- Preparation for migrating RedHat/CentOS cluster nodes to CoreOS/Fedora CoreOS
- Improved integration with the existing infrastructure

06/2016 – 10/2020

Hornbach Baumarkt

DevOps Engineer

Design, Deployment, and Operation of a PaaS Solution

The project aimed to identify and implement a Platform as a Service (PaaS) solution that would enable agile development teams to work fully autonomously, in alignment with existing Scrum processes, without dependencies on other teams such as network or infrastructure.

Requirements

**Docker, Kubernetes,
Ansible, RedHat,
Python, Java-Umfeld**

- Support for service-oriented architecture (SOA)
- Scalability
- Simple and standardized recovery processes
- Integration with the existing legacy environment featuring mixed infrastructure (Windows/Linux, physical/virtual)
- Facilitation of autonomous team workflows
- Capability to decouple additional services from the legacy infrastructure and migrate them onto the PaaS platform

Following the evaluation phase, the decision was made to build the PaaS solution based on Docker and Kubernetes. The deployment was carried out in close collaboration with the customer's Network, Infrastructure,

and Development teams.

After the platform was established, operational responsibilities were handed over to the Infrastructure team, which had received targeted training.

Responsibilities

- Evaluation of available open-source tools
- Platform design and deployment
- Configuration management
- Documentation creation
- Support for development teams
- Training and operational handover to the Infrastructure team

07/2019 – 12/2019

Hornbach Baumarkt

DevOps Engineer

**Active Directory,
Security Assertion
Markup Language
(SAML), LDAP, JSON
Web Token (JWT),
Ansible, PowerShell,
Kubernetes, GitLab,
Jenkins, ELK Stack,
Nexus Repository,
Microsoft Azure,
Google Cloud, Docker
Registry**

Unified and Comprehensive Authorization Concept

Over the years, several issues emerged in the area of permission management. There was no unified authorization concept; system administrators each requested separate sections in Active Directory to manage their systems independently. As a result, user permission assignments became cumbersome due to the lack of automation. Additionally, any reorganization or optimization of the organizational structure required significant effort because organization-specific permissions had to be adjusted accordingly.

An analysis of the current (AS-IS) and desired (TO-BE) states identified several key points. After multiple discussion rounds and collaboration with:

- System administrators
- Team leaders
- Active Directory administrators

requirements for the authorization concept were defined. Subsequently, the necessary tools were developed and deployed.

Requirements

- Support for all existing systems
- Integration with interfaces such as LDAP and SAML
- Project-based permissions
- Team leaders can define team structure (Team Roles)
- Team leaders can assign permissions to their team members for their projects
- Support for diverse system types (System Roles)
- Idempotency
- Extensibility

	Responsibilities • Analysis of AS-IS and TO-BE states • Research on possible solutions and approaches • Development and validation of the authorization concept • Tool development for implementation of the concept • Deployment and commissioning of the tool • Documentation and handover
09/2014 – 06/2016 Plus.line Systems Engineer Ansible, Cloudstack, KVM, Networking, RedHat	Design and Deployment of a Cloud Platform A cloud infrastructure can enable the use of more flexible as well as predefined (well-tested) solutions, increase independence from existing hardware assets, and thereby facilitate faster and higher-quality delivery of customer projects. For the implementation of this project, oVirt, CloudStack, and OpenStack were initially selected, deployed, and evaluated. The findings from the evaluation phase were compiled into a decision document and presented to the client. CloudStack was chosen for deployment due to its optimal fit with the requirements, offering a balance of flexibility and manageable complexity. In collaboration with network specialists, a production-ready platform based on CloudStack was developed. Typical end-user requirements were addressed through test scenarios, which were executed and evaluated. Additionally, disaster recovery tests were designed and conducted to proactively identify potential issues and prepare mitigation strategies. Subsequently, comprehensive documentation was created, and personnel were trained to ensure secure and stable platform operations. Responsibilities • Evaluation of available open-source tools • Platform design and deployment • Configuration management • Definition and execution of disaster recovery tests • Documentation creation
11/2015 – 02/2016 Deutsche Vermögensberatung Systems Engineer Puppet, Ruby, Shell, RedHat, Projektmanagement	Migration of Linux Systems As part of the project, over 400 Linux systems needed to be migrated (rebuilt) to the latest configuration management standards. A major challenge was not only mapping the systems using Puppet but also obtaining approval from business stakeholders and technical testers. To enable the rebuilds to be performed flexibly during off-peak hours (nights and weekends) with below-average system load, operations staff (24/7) received additional training. Leveraging strong DevOps expertise, complex production systems were

	modernized through collaboration between experienced software and systems engineers to ensure that potential issues - whether software-related or system configuration errors - could be identified and resolved rapidly. Following the migration, a stabilization phase was planned and executed to detect and address any new operational issues promptly. Responsibilities • System modeling with Puppet • Training of operations staff • Planning and coordination of migration activities • System support during and after migration
10/2013 – 05/2016 Deutsche Vermögensberatung Systems Engineer Puppet, Ruby, Shell, RedHat, KVM, Tomcat, Serverspec	Administration and Operation of the DVAG Platform The DVAG platform provides essential tools for financial advisors. Besides the production environment, the platform also includes testing, integration, and development environments. A wide range of technologies is utilized across the platform, which encompasses approximately 2,000 servers. One of the primary goals is to reduce administrative overhead by using the Puppet configuration management tool. The objectives include deploying the latest Puppet version and ensuring the quality and maintainability of the Puppet code. The Puppet code is tested through nightly builds executed and evaluated in a separate environment. Testing tasks have been automated using scripts, and server configurations are verified using the Serverspec tool. Responsibilities • Server administration with Puppet • Platform operation and maintenance • Test automation • Minimizing deployment downtime
07/2011 – 10/2013 General Motors Linux Systems Engineer RedHat, Xen, MySQL, Puppet, Shell, Perl	Operation of the GM World Platform The GM World Platform was designed to consolidate all markets (websites) of the GM brands (Chevrolet, Opel, Buick, Cadillac, etc.). The platform consists of multiple environments (DEV/ITL, PRE, PROD) and encompasses over 400 servers. Operational activities include, but are not limited to: • Implementation of changes • Incident resolution

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- Troubleshooting

All activities are conducted following the ITIL v3 framework.

The client is supported by a monitoring system and a 24/7 hotline, ensuring timely detection and resolution of issues around the clock.

Responsibilities

- Platform operation
 - Technical support and consulting for clients (Frankfurt, Detroit)
 - Performance analysis
 - Test team support (error analysis)
 - Monitoring
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02/2013 – 10/2013

General Motors

**Linux Systems
Engineer**

Shell, Perl, Python, Ant

Design and Implementation of Automated Deployments

Software deployments on the GM World Platform can take several hours due to their high complexity, typically requiring at least one administrator's full attention and intense focus.

The platform includes multiple environments supporting the software lifecycle: Development, Testing, and Production. Consequently, a release is installed multiple times across these environments (DEV/ITL, PRE, PROD) until all issues are resolved and the release can be deployed live.

The project goal was to automate these complex deployment processes to run without administrator intervention. Administrator involvement is therefore only necessary in case of errors, allowing a single administrator to oversee multiple deployments concurrently. Deployment duration is further reduced by executing deployments simultaneously across multiple components.

Responsibilities

- Reducing deployment time
 - Eliminating human error
 - Enabling concurrent deployments
 - Preparing for a growing number of markets and deployments
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01/2010 – 06/2011

Lufthansa Passage

**Project Manager,
Software Engineer**

**C/C++, Python, Perl,
Oracle, SQL, Linux,**

Maintenance and Enhancement of the Central Traffic Control System

The traffic control system—STORM (System Tool for Operation Rotation and Maintenance Requirements Traffic)—is used for planning, managing, and monitoring flights as well as ground events involving Lufthansa Passage aircraft.

As a real-time system, it compares the actual situation against the planned flight schedule, enabling users to take corrective or preventive actions to ensure punctuality, regularity, cost-efficiency, and safety.

Shell Scripting, XML	STORM offers a wide range of features, developed and tailored over the years specifically to meet the requirements of Lufthansa Passage. Additionally, it interfaces with more than 20 external systems to provide users with all necessary information. As one of the most critical applications for maintaining flight operations, the system must guarantee the highest possible availability. Responsibilities • Project management • Release planning • Resource planning • Maintenance and enhancement of the frontend application • Creation of documentation, specifications, and proposals • Test case design, execution, and analysis • Research
11/2007 – 06/2011 Lufthansa Passage DB-Administrator	Administration of Crew Management Systems (CMS) Databases The Crew Management Systems comprise over 15 Oracle databases (for both test and production environments) used by more than 22,000 employees. These databases contain a vast amount of personal data, interface with approximately 30 other systems, and are actively maintained by a team of 60 staff members. The critical databases operate in high-availability clusters to ensure maximum uptime, with failover times capped at under 5 minutes to maintain continuity in emergency situations. Responsibilities • Database administration • Creation and maintenance of documentation • Designing backup, restore, and reorganization procedures • Providing regular status reports for individual databases • Ensuring and monitoring high availability and data backups • Configuration and performance optimization • Development and improvement of operational concepts • Error analysis and storage requirement trend monitoring • Providing tuning recommendations for specific applications • Version planning and testing

09/2009 – 05/2010

Multiple Airlines

Web Developer

**Python, MySQL, SQL,
Javascript, Linux,
Shell Scripting,
Apache, Zope**

Development of a Reporting Web Portal for the European Emissions Trading System

The EU Emissions Trading System (EU ETS) is a market-based instrument of EU climate policy aimed at reducing greenhouse gas emissions at minimal economic cost. Airlines are required by this system to report all flights within a year that involve a European airport to the relevant European authority.

Due to the size of the airline and the complex EU requirements, generating the necessary reports involves significant effort and a complicated process.

The primary goal of the web application is to simplify the reporting process, provide a clear overview of the current status, and enable forecasting. Forecasting is particularly important as it allows early initiation of the purchase of required emission certificates, resulting in cost savings. Additionally, the system facilitates the generation of standardized reports that airlines must submit annually to national authorities.

Responsibilities

- Development and integration of the frontend into a web portal
- Coordination and synchronization with backend development
- Setup of the development environment
- Creation and execution of test cases
- Research focused on performance optimization

02/2007 – 12/2009

Lufthansa Passage

Software Engineer

**C/C++, Python, Perl,
Oracle, SQL, XML,
Linux, HP/UX, Shell
Scripting**

Maintenance and Enhancement of the Central Traffic Control System

The traffic control system - STORM (System Tool for Operation Rotation and Maintenance Requirements Traffic) - is used for planning, managing, and monitoring flights as well as ground operations of Lufthansa Passage aircraft.

As a real-time system, it presents the current operational status relative to the planned flight schedule and enables users to take corrective or preventive actions to ensure punctuality, regularity, cost-efficiency, and safety.

STORM includes a wide range of features tailored over the years specifically to meet the requirements of Lufthansa Passage. It also integrates with more than 20 external systems to provide users with all necessary information.

Due to its critical role in maintaining flight operations, the system demands the highest possible availability.

Responsibilities

- Maintenance and enhancement of the frontend application
- Maintenance and development of decision support components
- Workstation administration
- Specification development
- Support in executing and analyzing test cases
- Research

11/2007 – 11/2008**Lufthansa Passage****Software Engineer****C/C++, Oracle, SQL,
HP/UX, Linux****Porting of the Central Traffic Control System from HP/UX to Linux**

To identify the areas of the codebase suitable for performance optimization, a profiling analysis was conducted in preparation. Based on defined standard business use cases, the internal frontend execution times for individual functions were measured and evaluated. Following this analysis, optimization measures were defined in coordination with the client. One of these measures involved migrating the system from HP/UX to Linux.

The migration was necessary because the manufacturer of the existing workstations no longer supplied current models with four graphics outputs and, specifically, ceased support for such configurations. In contrast, Linux offered ongoing hardware support with modern, high-performance workstation options.

The goal of the migration was to enhance the application's performance, especially improving the speed of the GUI components.

Responsibilities

- Adapting the software code for Linux compatibility
 - Configuring Linux workstations to match the previous HP/UX workstation setup
 - Creating documentation and a setup manual for Linux workstations
 - Executing and analyzing test cases
 - Conducting research
-